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MIL-G-15342B
16 August 1971
SUPERSEDING
MIL-G-15342A
24 October 1951
(See 6.5)

MILITARY SPECIFICATION

GASKETS, METALLIC ASBESTOS, SPIRAL WOUND, FOR BOILERS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metallic-asbestos gaskets for handholes and manholes of shipboard and stationary boilers. Nuclear boilers are excluded.

1.2 Classification. Gaskets shall be of the following classes and series, as specified (see 6.1):

Class 1 - Boiler handhole.
Class 2 - Boiler manhole.

Series 600
Series 1500

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
QQ-S-766 - Steel Plates, Sheets, and Strip - Corrosion Resisting.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-up.

MILITARY
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL
FED-STD-151 - Metals, Test Methods.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

MILITARY
5000-S5101-841218 - Gasket, Boiler Handhole and Manhole, Steam Service.

FSC 5330

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, N.Y. 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 167 - Specification for Corrosion-Resisting Chromium-Nickel. Steel Plate, Sheet and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Metallic-asbestos gaskets furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Construction. Gaskets shall be composed of a single thickness of metal wound in a spiral shape, starting on the inside and working outward ply by ply to the desired size. Between the plies of metal shall be a cushion of asbestos filler material wound to conform with the metal strip. For class 1 gaskets, terminal spot welds on the gasket inside and outside ply shall be not less than three in number and spaced approximately 1/8 inch apart. For class 2 gaskets, terminal spot welds on the gasket inner ply shall be not less than three in number at the start of the ply. Additional spot welds shall be made so that spacing between spot welds is not greater than 3 inches. The outside welds shall consist of not less than three spot welds. Alternate methods to this construction shall be letter approved by the Naval Ship Engineering Center prior to bid submission, and accompany bid submission.

3.2.1 Materials.

3.2.1.1 Metal strip. The metal strip used in the gasket shall be one of the following materials:

(a) Corrosion resisting steel, AISI type 347, cold rolled and annealed, in accordance with ASTM A167; or QQ-S-766, type 347, cold rolled and annealed.
(b) Corrosion resisting steel, AISI type 316L, in accordance with ASTM A167; or QQ-S-766, type 316L.
(c) Corrosion resisting steel, AISI type 309S, in accordance with the applicable requirements of ASTM A167 except that type 309S shall be stabilized with columbium (and tantalum not exceeding 0.10 percent) and their combination to be not less than 10 times the carbon content; or QQ-S-766, type 309S, stabilized and combined as noted above.

3.2.1.2 Thickness. The thickness of the metal strip shall be 0.007 inch, plus or minus 0.0007 inch.

3.2.2 Asbestos filler.

3.2.2.1 Composition. The asbestos paper filler shall consist of chrysotile asbestos fiber with not more than 10 percent binder content of suitable rubber or paraffin.

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3.3 Gasket thickness tolerance. The thickness tolerance of the finished gasket as measured over the metal shall be plus or minus 0.005 inch.

3.4 Service conditions.

3.4.1 Service conditions and temperature ratings. Gaskets shall be rated as suitable for service at the following temperatures:

Class 1 - Handhole plate gaskets:
Series 600 - 850°F. Fahrenheit (F.)
Series 1500 - 950°F.
Class 2 - Manhole plate gaskets:
Series 600 - 500°F.
Series 1500 - 600°F.

3.4.2 Service condition of bearing surface. Gaskets shall be suitable for service in connection with flat smooth surfaces having finishes of spiral or concentric lay of 60 to 80 serrations per inch and a roughness height rating of 125.

3.5 Dimensions and tolerances.

3.5.1 Dimensions. Unless otherwise specified (see 6.1), class 1 and 2 metallic-asbestos gaskets shall conform to the dimensions shown as applicable to this specification on Dwg. 5000-S5101-841218. Class 2 gaskets shall be a true ellipse.

3.5.1.1 Number of plies. For class 1 metallic asbestos gaskets, the minimum number of plies of metal shall be seven, and for asbestos, four plies. Class 2 metallic asbestos gaskets shall contain not less than 42 plies of metal per inch of cross section of the bearing surface.

3.5.1.1.1 A ply shall consist of one 360-degree turn of the metal or asbestos strip. Gaskets shall be wound on a mandrel of the desired shape. After one complete revolution, the initial spot welds shall be made. Approximately midway during the next revolution, the asbestos filler strip shall be introduced. Winding shall continue to make the finished product contain the number of plies of asbestos required. A minimum of two plies of metal shall then be made and then the terminal spot welds.

3.5.2 Dimensional tolerances. Dimensional tolerances for class 1 and 2 metallic-asbestos gaskets shall be as shown as applicable to this specification on Drawing 5000-S5101-841218.

3.5.3 Compression load. The thickness of metallic-asbestos gaskets after compression load simulating service conditions shall be greater than the following:

(a) Initial gasket thickness of 0.175 inch. Thickness after compression 0.120 inch, plus or minus 0.005 inch.
(b) Initial gasket thickness of 0.125 inch. Thickness after compression 0.100 inch, plus or minus 0.005 inch.

3.6 Marking. Gasket shall be marked with a sturdy paper tag (of tag stock) securely attached thereto, which shall contain thereon the following information (see 6.1): The manufacturer's name; gasket size, including shape, thickness, inside and outside dimensions; type of metal and filler material; pressure and temperature rating, Federal stock number, and customer's order number. For class 1, at least one tag shall be applied to each 25 gaskets or less. For class 2 gaskets, each gasket shall be tagged.

3.7 Workmanship. The finished gaskets shall have no unnecessary breaks, creases or sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Qualification inspection.[1/] Qualification inspection of both classes of gaskets shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Manufacturers shall furnish a minimum of six representative samples for each class and series of gasket for qualification testing. A supplier may have his gaskets qualified for the service conditions specified by passing the test for the series concerned. Qualification examination and tests shall consist of the examination of 4.4 and tests of 4.5.

4.3 Sampling.

4.3.1 Lot. All gaskets of the same size and series manufactured from the same batch of material and presented at one time shall be considered a lot for purposes of examination and tests.

4.3.2 Sampling for examination. A random sample of gaskets shall be selected from each lot of material offered for inspection of visual and dimensional characteristics in accordance with the sampling procedures of MIL-STD-105, Special Inspection Level S-1. Acceptable Quality Level (AQL) shall be 1.5 percent defective.

4.3.3 Sampling for chemical analysis. A random sample of gaskets shall be selected from each lot in accordance with MIL-STD-105, Special Inspection Level S-3.

4.4 Visual and dimensional examination. Each of the sample gaskets selected in accordance with 4.3.2 shall be visually and dimensionally examined to verify compliance with this specification. Any gasket in the sample containing one or more defects shall not be offered for delivery, and if the number defective exceeds the acceptance number, this shall be cause for rejection of the lot represented by the sample. The number of plies shall be counted, a magnifying glass being used to insure accuracy of the count. Examination shall be conducted as specified in table I.

Table I - Classification of defects.

Categories	Defects
Critical	None defined.
Major:	
# 101	Class of gasket not or not as specified (see 6.1).
# 102	Rating of gasket not specified (pressure series) (see 6.1).
103	Number of metal plies less than specified minimum.
104	Number of asbestos plies less than that specified or plies less than a 360 degree turn each.
105	Spot welds (on inside and outside diameters) missing, not the required number, incorrectly spaced, or spot welds defective.
106	Thickness of metal strip not within specified maximum and minimum values, or metal strip not continuous (one piece).
107	Thickness of finished gasket not within specified tolerances.
108	Dimensioning of gasket not as specified in contract or on standard drawing.
# 109	Unnecessary breaks, creases, or sharp edges on finished gaskets.
# 110	Identification markings missing on tags, illegible, incorrect, or not as specified.
# 111	Packaging or packing not as specified.

4.4.1 Quality conformance test. Each of the sample gaskets selected in accordance with 4.3.3 shall be analyzed in accordance with method 111 of FED-STD-151 to determine compliance with 3.2.1.1. If any sample fails to conform to 3.2.1.1, this shall be cause for rejection of the entire lot.

4.5 Test procedures.

4.5.1 Chemical analysis.

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.4).

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4.5.1.1 Qualification test. Sample gaskets specified in 4.3.3 shall be analyzed by method 111 of FED-STD-151 to determine compliance with 3.2.1.1. Chemical analysis shall also be made of the asbestos filler material during qualification testing to determine compliance with 3.2.2.1.

4.5.2 Compression load test (for class 1 gaskets). Tests shall be conducted in a handhole gasket test apparatus under operating conditions similar to those for which the gasket is intended. The apparatus shall consist of a header or series of headers containing handhole openings in which test gaskets will be installed. Bearing surfaces of joint shall have a semi-fine tool finish consisting of spiral or concentric lay of 60 to 80 serrations per inch and a roughness height rating of 125. A hydrostatic test with water at 70°F. and at 1.25 times the pressure for which the gaskets are intended shall be applied to the header and handhole plate nuts tightened to eliminate leakage. Water shall then be drained from the header and steam admitted. Temperature shall gradually be increased to that for which the gasket is to be qualified. Steam shall be admitted for a minimum of four hours after which the steam supply valve shall be closed and the drain valve opened and the test apparatus allowed to cool overnight. A hydrostatic test shall be made every fifth day at 1.25 times operating pressure with water at 70°F. Test shall be conducted for a minimum of one hundred hours operation at the service conditions. The joints shall then be broken, the gaskets visually and dimensionally examined, and the thickness measured. If within the thickness limitation outlined in 3.5.3 and in satisfactory condition, the gaskets shall be considered to have passed this test.

4.5.3 Compression load test for class 2 gaskets. Tests shall be conducted in the joints of a gasket test apparatus under conditions simulating those encountered in service. Bearing surfaces of joint shall have a semi-fine tool finish consisting of spiral or concentric lay of 60 to 80 serrations per inch and a roughness height rating of 125. The gasket shall be installed in the test apparatus and the joint assembled. A hydrostatic test with water at 70°F. and at 1.25 times service condition pressure shall be made to prove tightness of the joint. Steam at service condition pressure shall be established in the drum and the water level maintained so that the upper half of the test gasket is exposed to steam and the lower half to water. The water level and steam conditions shall be maintained for an elapsed time of one hundred hours. After releasing the pressure, the joint shall be broken, the gaskets visually and dimensionally examined, and the thickness measured. If within the thickness limitation outlined in 3.5.3, and in satisfactory condition, the gaskets shall be considered to have passed this test.

4.6 Possible test failures. Possible test failures are defined as follows:

(a) Steel strip not as specified.
(b) Asbestos filler not as specified.
(c) Gasket buckled excessively on inner periphery to the extent that metal plies are separated.
(d) Welding failed.
(e) Gasket thickness recovered less than that specified, after release of compression test load.

4.7 Comparison inspection. Comparison inspection shall be conducted to verify continuing quality of the approved product and compliance with the specification requirements. These tests will be on samples representing those series and sizes which were tested for qualification and approved, and will be conducted on six samples of each series and size, supplied by the manufacturer at no cost to the Government. Comparison inspection is not required more often than every two years unless the inspector has reason to consider it necessary. Comparison examination and tests shall be those specified in 4.4 and 4.5 plus any others specifically requested by the inspector.

4.8 Inspection of preparation for delivery. Sample packages and packs and the inspection of the packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.2.)

5.1 Packaging. Packaging shall be level A or level C, as specified (see 6.1).

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5.1.1 Level A. Gaskets, quantity as specified (see 6.1), shall be packaged in boxes conforming to PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676, at the option of the supplier. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.1.1.1 Bundling. Gaskets shall be bundled in quantities, tied, and wrapped in a non-corrosive, dust protective opaque or transparent paper or film as follows:

Class 1 - 25 gaskets per bundle
Class 2 - 12 gaskets per bundle

The wrapper shall be secured with pressure sensitive tape or heat sealed.

5.1.2 Level C. Gaskets shall be packaged in a manner to afford protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and immediate use. The supplier's normal packaging method may be used when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Gaskets, packaged as specified (see 6.1), shall be packed in fiberboard boxes conforming to class weather-resistant of PPP-B-636. Alternately, gaskets may be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-576	Class 2
PPP-B-585	Class 3
PPP-B-591	Class II
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-640	Class 2
PPP-B-665	Weather Resistant

Shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Caseliners for class weather resistant fiberboard boxes conforming to PPP-B-636 may be omitted provided all center and edge seams and manufacturer's joints are waterproofed in accordance with the appendix to PPP-B-636. Boxes shall be closed, strapped or banded in accordance with the applicable box specification or appendix thereto (except that fiberboard boxes shall be reinforced in lieu of steel strapping, with pressure-sensitive, reinforced filament tape and applied in accordance with the appendix to the applicable fiberboard box.) The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.2 Level B. Gaskets, packaged as specified (see 6.1), shall be packed in fiberboard boxes having a minimum bursting strength of 275 p.s.i., conforming to PPP-B-636, class domestic. Alternatively, gaskets may be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-576	Class 1
PPP-B-585	Class 1 or 2 use
PPP-B-591	Class I
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-640	Class 1
PPP-B-665	Weather resistant

5.2.2.1 Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds. Not more than one size or type of gaskets shall be packed in a single container.

5.2.3 Level C. Gaskets of like series and size, packaged as specified (see 6.1), shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

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5.2.4 Filling of voids. Voids or cavities in the shipping containers shall be filled with cushioning materials to prevent shifting of contents during handling, shipment and storage. The use of loose excelsior, shredded newspaper and similar materials which have not been refined for non-corrosiveness shall not be used unless these materials are enclosed in a sealed waterproof barrier material. Use of expanded, loose-fill, polystyrene is permitted when approved (see 6.1) by the procuring activity.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.1), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and series of gasket (see 1.2).
(c) Full description such as where used in boiler, inside and outside dimensions, thickness, bearing surface, etc., or line item number from standard drawing (see 3.5 and 3.6).
(d) Level of packaging and packing (see 5.1 and 5.2).
(e) Quantity per unit package (see 5.1.1).
(f) When use of expanded, loose-fill polystyrene is approved (see 5.2.4).
(g) Special marking, if required (see 5.3).

6.2 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 15342 whether or not such products have actually been so listed by that date. The attention of the supplier is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4).

6.4 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Custodians:
Army - MR
Navy - SH
Air Force - 82
Review activities:
Army - MR
Navy - SH, SA
Air Force - 82
DSA - IS
User activities:
Army - ME
Navy - YD

Preparing activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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